

# Water Jug Problem Using Python

## Aim

To write a Python program to solve the **Water Jug Problem** using the **Breadth-First Search (BFS)** algorithm and determine the sequence of steps required to obtain the target amount of water.

## Algorithm

1. Start the program.
2. Initialize both jugs as empty (0, 0).
3. Create a queue to store the states and a set to keep track of visited states.
4. Remove the front state from the queue.
5. If the current state contains the target amount of water, display "Target reached" and stop.
6. Otherwise, generate all possible next states:
  - Fill Jug 1.
  - Fill Jug 2.
  - Empty Jug 1.
  - Empty Jug 2.
  - Pour water from Jug 1 to Jug 2.
  - Pour water from Jug 2 to Jug 1.
7. Add the unvisited states to the queue.
8. Repeat Steps 4–7 until the target is reached.
9. Stop the program.

## Python Code

```
from collections import deque

def water_jug(x, y, target):
    visited = set()
    q = deque([(0, 0)])
    while q:
        a, b = q.popleft()
        if (a, b) in visited:
            continue
        visited.add((a, b))
        print(a, b)
        if a == target or b == target:
            print("Target reached")
            return
```

```

        q.extend([
            (x, b),
            (a, y),
            (0, b),
            (a, 0),
            (max(0, a - (y - b)), min(y, b
+ a)),
            (min(x, a + b), max(0, b - (x
- a)))
        ])
    water_jug(4, 3, 2)

```

```

EXP 3(Water Jug Problem).py > water_jug
1  from collections import deque
2  def water_jug(x, y, target):
3      visited = set()
4      q = deque([(0, 0)])
5      while q:
6          a, b = q.popleft()
7          if (a, b) in visited:
8              continue
9          visited.add((a, b))
10         print(a, b)
11         if a == target or b == target:
12             print("Target reached")
13             return
14         q.extend([
15             (x, b),
16             (a, y),
17             (0, b),
18             (a, 0),
19             (max(0, a-(y-b)), min(y, b+a)),
20             (min(x, a+b), max(0, b-(x-a)))
21         ])
22     water_jug(4, 3, 2)

```

PROBLEMS OUTPUT DEBUG [Open file in editor \(ctrl + click\)](#)

```

PS E:\OPEN CV\AI Lab> & C:\Users\user\AppData\Local\Programs\Python\Python310\python.exe
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
Target reached

```